Jiachi ZHANG

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EDUCATION

South China University of Technology 09/2023 - 06/2027

***Qualification*:** Bachelor of Engineering

***Major*:** Information Engineering

***Average Score*:** 84.47/100

***Core Courses*:** *Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++*

SUMMER STUDY PROGRAM

École Polytechnique de l'Université de Nantes, France 07/2025

***Core Courses*:** *Français, théorie de l'information*

INTERNSHIP EXPERIENCE

The Chinese University of Hong Kong, Shenzhen 08/2026 - Present

Position: *Visiting Research Intern, Professor Ming Li's Research Group*

- Assisting with research on multimodal behavioural assessment for autistic children, with an initial focus on multimodal behavioural signals.

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences 10/2025 - Present

Position: *Visiting Research Intern, Assistant Professor Wenqi Fang's Research Group*

- Carried out independent research by identifying research gaps, developing research hypotheses, designing experimental protocols, and conducting hands-on experiments and data analysis.
- Compiled academic materials by reviewing relevant literature and drafting concise and structured experimental summaries.
- Authored and revised the thesis, ensuring a clear, logically rigorous scientific narrative grounded in experimental results.

RESEARCH EXPERIENCE

Gain-Aware Prediction-Space Recursive Controller for Lightweight Polyp Segmentation

Accepted at ICONIP 2026 | **Jiachi Zhang** et al. **(First Author)**

• Formulated segmentation refinement as prediction-space recursive correction with only 189 parameters and negligible GMACs overhead.

• Designed a three-stage recursive controller: Evidence Encoding, Gain-Aware State Update, State-Guided Residual Correction.

• Evaluated across 7 lightweight backbones on 4 datasets, consistently outperforming training-side baselines (+LL/+DS/+Mutation/+LoMix) and +Harmonizing.

RIGS-Refiner: Risk-Guided Recursive Refinement in Prediction Space for Polyp Segmentation

Accepted at APSIPA ASC 2026 | **Jiachi Zhang** et al. **(First Author)**

• Proposed a lightweight post-refinement plugin with 519 parameters combining image priors, prediction cues, risk-guided update, and residual correction.

• Tested on PraNet and SegFormer-B0 across 4 datasets, achieving favorable efficiency–accuracy trade-offs while using approximately 1/2,600 as many parameters as CascadePSP.

Additional Research Experience

- Developed FireflyNet, an ultra-lightweight recursive segmentation framework with deployment-oriented evaluation (latency, memory, heterogeneous runtime benchmarks).
- Investigated knowledge distillation with few-shot transfer learning for OCT segmentation targeting edge deployment.

- Explored stain normalization (color clustering + transfer) and YOLO11-based detection or glomerular pathological image analysis.

PROJECT EXPERIENCE

AI-Powered Smartwatch Health Management System (Course Project, Team Lead) 05/2026

- Built an end-to-end four-layer system spanning ESP32 firmware (C/C++, FreeRTOS), MQTT-based IoT communication, Flutter web/mobile client, and Node.js backend with LLM-driven health analysis
- Designed MQTT topic architecture, bidirectional device control protocol, and a unified multi-source domain model supporting multi-user health profiles
- Implemented deterministic rule engine for health alerts with LLM-generated trend summaries and a human-in-the-loop confirmation mechanism
- Tech stack: C/C++, ESP-IDF, FreeRTOS, Flutter/Dart, Node.js, MQTT

FPGA-Based Shooting Confrontation Game 03/2026

- Developed a dual-mode FPGA shooting game supporting 1280×800@60fps HDMI output, infrared control and real-time score display.
- Designed HDMI display drivers and SDRAM caching logic, and implemented AABB-based collision detection for game objects.
- Used finite state machines to control seven game states and ensure synchronization between peripherals and display modules.
- Completed project validation and received the Excellent Course Design award for outstanding academic performance.

EXTRACURRICULAR EXPERIENCE

President, Art Troupe, Youth League Committee 11/2024 - 11/2025

- Participated in the production and singing of the department song, and performed in the college band on various occasions.
- Managed and coordinated internal departmental affairs, and took charge of the college graduation gala as a key organizer for multiple times.

Study Representative, Class Committee 09/2023 - 10/2025

- Collected and distributed daily assignments, and completed the liaison work for academic affairs.
- Communicated actively between teachers and classmates, and released official university notifications in a timely manner.

HONORS & AWARDS

3rd Prize Scholarship, South China University of Technology 12/2025
Merit Student, South China University of Technology 12/2025
Advanced Individual in Student Work, South China University of Technology 09/2025
Active Participant in Student Work, South China University of Technology 09/2025
Outstanding League Cadre, South China University of Technology 05/2025

ADDITIONAL SKILLS

Language Ability: French (basic), English (fluent, can be used as a studying language)
Programming & Development: Python, PyTorch, Keil, MATLAB, Vivado